JAX-SKLEARN (xlearn) — SYSTEM ARCHITECTURE

drop-in sklearn · files: xlearn/__init__.py + xlearn/_jax/*.py · github.com/chenxingqiang/jax-sklearn

USER CODE

import xlearn as sklearn
public API

LinearRegression / KMeans / PCA
xlearn.linear_model · cluster · decomposition

set_config / config_context
xlearn/_jax/_config.py

PACKAGE INIT xlearn/__init__.py

__getattr__(name)
lazy submodule import

inject

_JAX_ENABLED
import xlearn._jax

_auto_jax_accelerate_module
replace estimator classes

AcceleratorRegistry
xlearn/_jax/_accelerator.py

PROXY xlearn/_jax/_proxy.py

create_intelligent_proxy
wrap estimator type

EstimatorProxy.__init__
_create_implementation()

JAX

fallback

JAX path
create_accelerated_estimator

sklearn fallback
original_class(*args)

JAX KERNELS xlearn/_jax/

_universal_jax.py
generic mixins JIT/vmap

_accelerator.py
dispatch + registry

_data_conversion.py
NumPy ↔ JAX

_gp.py
Gaussian process

_mlp.py
neural_network

_svm.py
SVM

_tree.py
trees

CONFIG / HEURISTICS xlearn/_jax/_config.py

enable_jax
on/off

jax_platform
cpu | gpu | tpu | metal

LinearRegression
min_complexity 1e8

KMeans
min_complexity 1e6

PCA
min_complexity 1e7

HARDWARE (JAX backends)

CPU jaxlib
default

CUDA jax[gpu]
NVIDIA

TPU jax[tpu]
Google Cloud

Metal jax-metal
Apple experimental

Injection is lazy: xlearn.__getattr__ → proxy class in module → fit() chooses JAX or sklearn